

Contents

1	Causality	1
1.1	Association	1
1.1.1	Simplification using directed contractions	2
1.1.2	Statistical influence	2
1.2	Intervention	3
1.2.1	Do calculus	4
1.2.2	Intervention Queries	5
1.2.3	Simplifications	6
1.3	Counterfactuals	9
1.3.1	Response Variables	10
1.3.2	Counterfactual Twinned Network	12
1.4	Learning	13

Chapter 1

Causality

Abstract We here describe how to represent Causal AI models in the tntreason formalism. In particular, we will extend the Bayesian network formalism to capture intervention and counterfactual queries.

We find tensor network models for the different rungs of the *ladder of causation* [?]. The notes are organized to follow the three rungs of the ladder:

- **Rung 1: Association** – captured by Bayesian Networks
- **Rung 2: Intervention** – captured by do-calculus, for which we introduce intervention variables
- **Rung 3: Counterfactuals** – captured by functional causal models (see Chapter 21 in [?]) involving response variables

Following the formalism of Chapter 21 in [?]:

- Mutation of the Bayesian network is captured by contractions with one-hot encodings to do-variables D_k .
- Intervention queries are formalized by contractions with one-hot encodings to all do-variables
- Response variables are formalized by selection variables
- Counterfactual queries are modeled as intervention queries on counterfactual twin networks (see [?, Chapter 21])

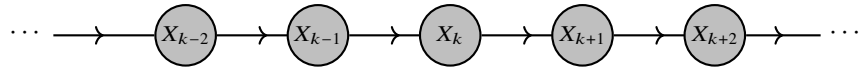
1.1 Association

See Chapter ??: Bayesian Networks are tensor networks and probabilistic queries can be answered by their contractions. We here provide more theory on query simplification and d-separation, to prepare for the study of intervention mechanisms.

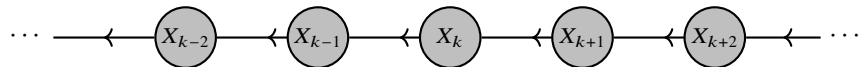
Example 1.1 (Markov chain)

Markov chains can be parametrized in two directions:

a) Past-to-future representation:



b) Future-to-past representation:



Both parametrizations represent the same distribution, and encode the same conditional independence statements. However, intuitively the future is influenced by the past, so we would prefer the parametrization a). We will observe in the following (see Example 1.3) how expanding the model through intervention mechanisms will provide us a method to distinguish between both parametrizations.

1.1.1 Simplification using directed contractions

Lemma 1.1 Let $X_{[d]}$ be an enumeration of the variables respecting a topological order on the directed acyclic hypergraph $\mathcal{G}$. For any $\tilde{k} < k$ we build the sets $[k] \setminus [\tilde{k}] = \{\tilde{k}, \tilde{k} + 1, \dots, k - 1\}$ and $\mathcal{U} = \left(\bigcup_{l \in [k] \setminus [\tilde{k}]} \text{Pa}(l) \right) \cap [\tilde{k}]$. Then

$$\mathbb{P}[X_{[k] \setminus [\tilde{k}]} | X_{\mathcal{U}}] = \langle \{ \mathbb{P}[X_l | X_{\text{Pa}(l)}] : l \in [k] \setminus [\tilde{k}] \} \rangle_{X_{[k] \setminus [\tilde{k}]}, X_{\mathcal{U}}}$$

Proof. By construction we have for $l \in [k] \setminus [\tilde{k}]$ the independencies

$$\left(X_l \perp X_{[l] \setminus ([\tilde{k}] \cup \text{Pa}(l))}, X_{\mathcal{U} \setminus \text{Pa}(l)} \right) \Big| X_{\text{Pa}(l)}$$

and therefore

$$\mathbb{P}[X_l | X_{\text{Pa}(l)}] \otimes \mathbb{I}[X_{[l] \setminus ([\tilde{k}] \cup \text{Pa}(l))}] \otimes \mathbb{I}[X_{\mathcal{U} \setminus \text{Pa}(l)}] = \mathbb{P}[X_l | X_{([l] \setminus [\tilde{k}]) \cup \mathcal{U}}].$$

It follows that

$$\langle \{ \mathbb{P}[X_l | X_{\text{Pa}(l)}] : l \in [k] \setminus [\tilde{k}] \} \rangle_{X_{[k] \setminus [\tilde{k}]}, X_{\mathcal{U}}} = \langle \{ \mathbb{P}[X_l | X_{([l] \setminus [\tilde{k}]) \cup \mathcal{U}}] : l \in [k] \setminus [\tilde{k}] \} \rangle_{X_{[k] \setminus [\tilde{k}]}, X_{\mathcal{U}}} = \mathbb{P}[X_{[k] \setminus [\tilde{k}]} | X_{\mathcal{U}}]$$

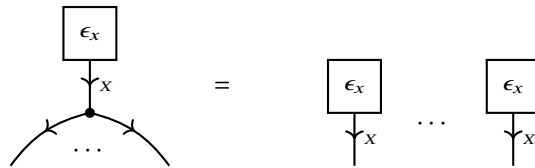
Here we used in the last equation the chain decomposition on the conditioned distribution $\mathbb{P}[X_{[k] \setminus [\tilde{k}]} | X_{\mathcal{U}}]$. $\square$

1.1.2 Statistical influence

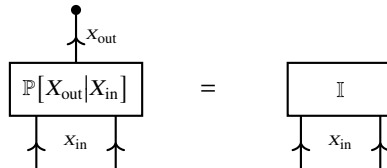
Let us investigate, how the directionality in a Bayesian network hypergraph can be interpreted as statistical influence between variables. To be more precise, we study which cores need to be contracted to answer probabilistic queries, and in such way contribute a statistical influence to a query.

When conditioning on some variables, and marginalizing others, we can simplify the Bayesian network. There are two simplification mechanisms:

- (i) When having a basis vector to a closed variable, we can remove the variable and replace all tensors affected by the variable with contractions with a copy of the basis vector. That is, we apply the contraction identity:

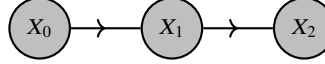


- (ii) When a closed variable appears only as an outgoing variable, we can replace the corresponding tensor with the trivial tensor. This corresponds with the contraction identity:



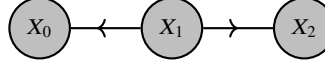
When having a path through the directed hypergraph (which itself does not necessarily respect the directionality), we investigate whether it will be a path in the simplified hypergraph. Consiering a node X_1 and its neighbors X_0, X_1 in a path, there are three different situations [?]:

- **Chains:** X_1 appears as outgoing and incoming variable.



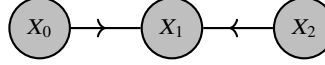
When conditioned on X_1 , the path closes by rule (i).

- **Forks:** X_1 appears only as outgoing variable.



When conditioned on X_1 , the path also closes by rule (i).

- **Colliders:** X_1 appears only as incoming variable.



When X_1 is marginalized out and not conditioned on, simplification rule (ii) disconnects the path. When X_1 is conditioned on, the path is still active by the tensor $\mathbb{P}[X_1 = x_1 | X_0, X_2, X_{\text{Pa}(1) \setminus \{0,2\}}]$

We use this intuition to define active trails and d-separation, oriented on [?, Def. 3.6 and Def. 3.7].

Definition 1.1 (Active trails and d-separation) A trail in a directed hypergraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is a enumerated subset of variables $X_{[r]}$ such that for each $s \in [r-1]$ there is a hyperedge $e = (e^{\text{in}}, e^{\text{out}})$ with $X_s \in e^{\text{in}}$ and $X_s \in e^{\text{out}}$ or $X_s \in e^{\text{out}}$ and $X_s \in e^{\text{in}}$. We say that a trail in $\mathcal{G}$ is blocked giving a set C , if any collider has a node in C or in $\text{Desc}(C)$ in the middle, and no chain or fork has a node in C in the middle. Two subsets A and B are said to be d-separated given C , if any trail from A to B is blocked by C .

To show the utility of this definition, let us consider two sets of variables A, B d-separated by C .

Theorem 1.1 *If and only if A, B d-separated by C then for any Bayesian network on $\mathcal{G}$*

$$(X_A \perp X_B) \mid X_C.$$

Proof. Let us first show, that whenever we have A, B which are d-separated by C and a Bayesian network on $\mathcal{G}$, then $(X_A \perp X_B) \mid X_C$ with respect to that Bayesian network. By performing the simplification rules iteratively on the contraction

$$\mathbb{P}[X_{A \cup B} | X_C = x_C]$$

and observing that the tensor network decomposes into a tensor product

$$\mathbb{P}[X_{A \cup B} | X_C = x_C] = \mathbb{P}[X_A | X_C = x_C] \otimes \mathbb{P}[X_B | X_C = x_C].$$

To show the contrary, it is enough construct a Bayesian network not satisfying this independence statement once A, B are not d-separated by C . We refer to Thm. 3.4 in [?] for such a construction. $\square$

We already characterized in Chapter ?? Bayesian networks on a hypergraph as those distributions satisfying the independencies

$$(X_k \perp X_{[k] \setminus \text{Pa}(k)}) \mid X_{\text{Pa}(k)}$$

with respect to a topological order on $\mathcal{G}$.

1.2 Intervention

So far, there are several hypergraphs to model a probability distributions by a Bayesian Network. We now want to understand the directed hyperedges as causal relationships between the variables, beyond encoding of conditional independencies (see Example 1.2). To this end, we model the effects of interventions, which will distinguish between multiple Bayesian Networks modeling the same joint distribution.

Example 1.2 (Patient treatment outcome) We model a relationship about the treatment of a sick patient and the outcome of the treatment. Whether a treatment is conducted is modelled by the binary variable X_T and whether the patient improves after the treatment by the variable X_O .

There are in general two possibilities of representing the joint distribution of X_T and X_O by a Bayesian Network:

- Treatment directs to outcome:

$$\mathcal{G} = (\{X_T, X_O\}, \{(\emptyset, X_T), (\{X_T\}, \{X_O\})\})$$

- Outcome directs to treatment:

$$\mathcal{G} = (\{X_T, X_O\}, \{(\emptyset, X_T), (\{X_O\}, \{X_T\})\})$$

In the typical intuition, we expect the treatment to have a causal effect on the outcome and not the other way round. From a Bayesian Network perspective, both models are however equivalent and represent the same joint distribution.

Interventions on Bayesian Networks are modeled by the do-calculus [?], which we capture here based on do-variables.

1.2.1 Do calculus

We introduce the do-variables D_k taking values in $[m_k + 1]$ for each variable X_k of dimension m_k , which model the effects of interventions. The states are interpreted as

- $d_k \in [m_k]$: An intervention setting X_k to the value d_k
- $d_k = m_k$: No intervention on X_k

Definition 1.2 The causally augmented conditional probability core is defined as

$$\begin{aligned} \mathbb{P}^c [X_k | D_k, X_{\text{Pa}(k)}] &= \mathbb{P} [X_k | X_{\text{Pa}(k)}] \otimes \epsilon_{m_k} [D_k] \\ &+ \sum_{d_k \in [m_k]} \epsilon_{d_k} [X_k] \otimes \epsilon_{d_k} [D_k] \otimes \mathbb{I} [X_{\text{Pa}(k)}] . \end{aligned}$$

The causally augmented Bayesian Network is the tensor network

$$\langle \{ \mathbb{P}^c [X_k | D_k, X_{\text{Pa}(k)}] : k \in [d] \} \rangle_{[X_{[d]}, D_{[d]}]} .$$

The Bayesian Networks investigated so far are retrieved by contractions with $\epsilon_{m_k} [D_k]$. This corresponds with the situation, where no interventions are performed on any variable.

Lemma 1.2 For any Bayesian Network $\mathbb{P} [X_{[d]}]$ on a directed acyclic hypergraph $\mathcal{G} = ([d], \mathcal{E})$ we have

$$\mathbb{P} [X_{[d]}] = \langle \{ \mathbb{P}^c [X_k | D_k, X_{\text{Pa}(k)}] : k \in [d] \} \cup \{ \epsilon_{m_k} [D_k] : k \in [d] \} \rangle_{[X_{[d]}]} .$$

Proof. By construction we have

$$\mathbb{P}^c [X_k | D_k = m_k, X_{\text{Pa}(k)}] = \mathbb{P} [X_k | X_{\text{Pa}(k)}] .$$

Performing this on each causally augmented conditional distribution retrieves thus the original Bayesian Network. $\square$

Adding the do-variables comes with a moderate increase of the bas+ rank. We provide the following bound.

Lemma 1.3 For any conditional probability distribution we have

$$\text{rank}^{\text{bas}} (\mathbb{P}^c [X_k | D_k, X_{\text{Pa}(k)}]) \leq \mathbb{P} [X_k | D_k, X_{\text{Pa}(k)}]$$

Example 1.3 (Intervention on a Markov Chain)

Let us recall the ambiguity of parameterizing a Markov Chain in Example 1.1, where we observed that both past to future and future to past representations result in Bayesian Networks of similar sparsity. Adding do variables to any of these parametrization will result in different interventions (see Figure 1.1) and thus resolves the ambiguity.

Intervening on a variable X_k and putting it into the state x_k is done by contracting with

$$\epsilon_{x_k} [D_k] .$$

We use that since $x_k \in \mathbb{R}^{m_k}$

$$\langle \mathbb{P}^c [X_k | X_{k-1}, D_k], \epsilon_{x_k} [D_k] \rangle_{[X_k, X_{k-1}]} = \mathbb{I} [X_{k-1}] \otimes \epsilon_{x_k} [X_k]$$

and have that the intervention effectively restarts the Markov Chain at the variable X_k (see Figure 1.3a). When we choose the causal augmentation of the future-to-past representation instead, the effect would be differently since the restart would be towards the past (see Figure 1.3b).

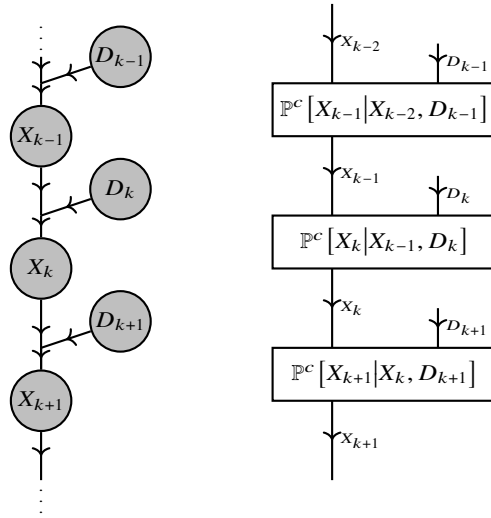


Fig. 1.1 Extension of a Markov Chain in the past-to-future representation by do variables.

1.2.2 Intervention Queries

In a Bayesian Network, when intervening on a variable X_v , the corresponding conditional probability gets trivialized. The probability tensor is then captured by

$$\mathbb{P} [X_{\mathcal{U}} | \text{do} (X_{\mathcal{W}} = x_{\mathcal{W}})] = \left\langle \{ \mathbb{P} [X_v | X_{\text{Pa}(v)}] : v \notin \mathcal{W} \} \cup \left\{ \frac{1}{m_v} \mathbb{I} [X_v, X_{\text{Pa}(v)}] : v \in \mathcal{W} \right\} \cup \{ \epsilon_{x_v} [X_v] : v \in \mathcal{W} \} \right\rangle_{[X_{\mathcal{U}}]} .$$

Since $\frac{1}{m_v} \mathbb{I} [X_v, X_{\text{Pa}(v)}]$ is directed with $X_{\text{Pa}(v)}$ incoming and X_v outgoing, the partition function of the network stays 1 and the normalization is captured by the contraction.

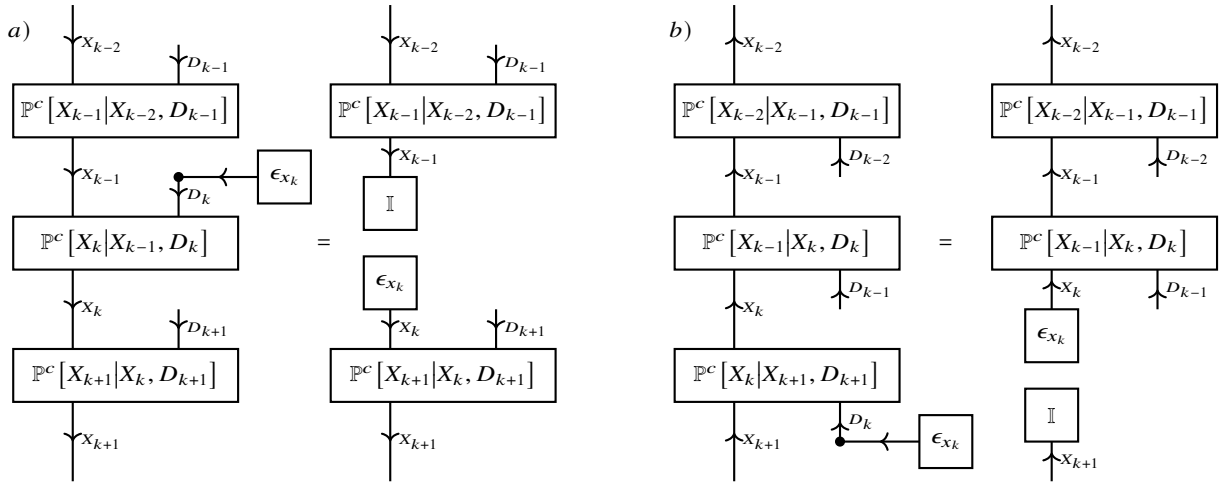


Fig. 1.2 Effect of intervention on a variable D_k : Restarting the Markov chain at the variable X_k in state x_k . The effect differs between the past-to-future parameterization a) and the future-to-past parameterization b).

1.2.3 Simplifications

In general, when all variables are observable, intervention queries can be expressed by a collection of conditional queries (since any cpd of a Bayesian network is a conditional query, and we contract all except those with interventions on the head variables). In realistic scenarios, however, not all variables are observable (for example, the smoking gene has been only assumed, but no observations have been made). We are interested in situations, where intervention queries can be answered by conditional queries of observable variables. We can prove them in the tensor network formalism based on network separations, which contribution to contractions are scalar multiplications, which therefore are dropped in normalizations.

An example for a simplification is given by the randomized controlled trial.

Example 1.4 (Randomized controlled trial) The randomized controlled trial is a joint distribution of variables

- Confounding variable X_C summarizing other factors affecting the outcome.
- Treatment assignment X_T with values in $\{0, 1\}$. Here, 1 indicates assignment to the treatment group and 0 to the control group. The experiment is conducted such that the treatment variable is independent of the confounding variable X_C .
- Outcome X_O .

We want to estimate the causal effect of the treatment on the outcome, namely calculate the intervention query

$$\mathbb{P}[X_O | D_T = 1].$$

By design of the randomized control, the confounding variable has no causal effect on X_T . Thus, we have

$$\mathbb{P}[X_O | D_T = 1] = \mathbb{P}[X_O | X_T = 1]$$

that is, the intervention query is equal to the conditional probability (see Figure 1.3).

If the treatment variable would depend on the confounder (see Figure 1.4), the effect of the intervention can not be identified based on conditional probabilities, when not measuring the confounder X_C .

Compared to the conditional queries studied in Sect. 1.1, interventions create further simplification possibilities.

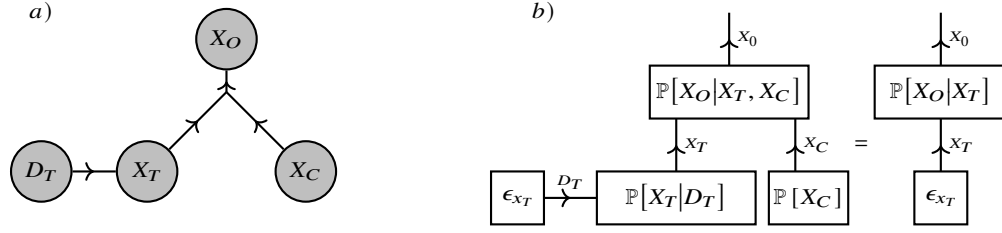


Fig. 1.3 Model of a Randomized Controlled Trial: The confounding variable X_C summarizes all influenced on the outcome X_O other than the treatment variable X_T . a) b) Effect of intervening on the treatment variable $x_T \in \{0, 1\}$

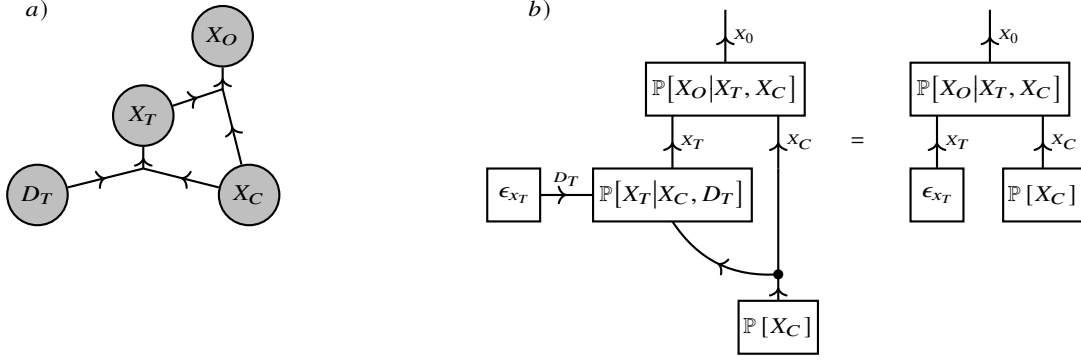


Fig. 1.4 Effect when the confounder has a causal influence on the treatment: The causal effect is not identifiable when not measuring the confounding variable.

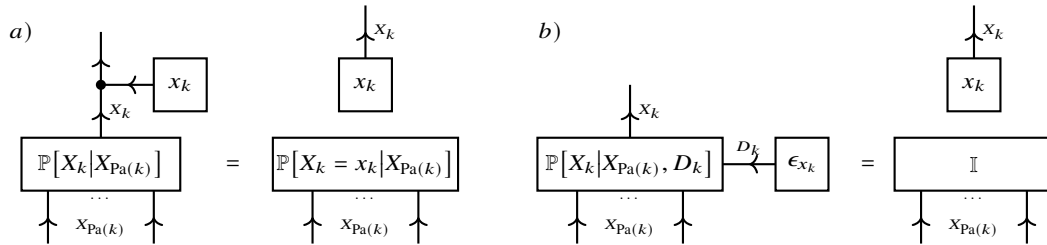


Fig. 1.5 Comparison of conditioning a) and intervention b) on a variable X_k . When intervening, a trivial tensor over the parent variables is left, which can be exploited for further query simplifications.

1.2.3.1 Backdoor Criterion

More general than in the randomized controlled trial we have to deal with confounding variables. The Backdoor Criterion and the Frontdoor Criterion exploit mediator variables X_M , which are also observable and block certain causal paths to unobserved confounding variables X_C .

Definition 1.3 Let there be a directed acyclic hypergraph, such that X_T is topologically ordered before X_O . A backdoor path is a path through the hypergraph, such that the first and the second node in the path are in the outgoing and the incoming nodes of a hyperedge. If every backdoor path from X_T to X_O is blocked by a variable in X_M , and X_M does not contain a descendant of X_T , then we say X_M satisfies the backdoor criterion with respect to X_T and X_O .

See Figure 1.6 for a typical hypergraph where X_M satisfies the backdoor criterion.

Lemma 1.4 When X_M satisfies the backdoor criterion for X_T and X_O , and for some $x_M \in [m_M], x_T \in [m_T]$ we have $\mathbb{P}[X_T = x_T, X_M = x_M \neq 0]$ then

$$\mathbb{P}[X_O|X_T = x_T, X_M = x_M] = \mathbb{P}^c[X_O|D_T = x_T, X_M = x_M].$$

Proof. We denote $\mathcal{U}$ as the set of nodes appearing in backdoor trails from X_T to X_M and those variables having a variable in X_T as a descendant. Since any backdoor path is blocked by X_M , we can isolate the contraction of the nodes in $\mathcal{U}$ by simplifying the contraction as

$$\begin{aligned} \mathbb{P}^c [X_O, D_T = x_T, X_M = x_M] &= \langle \{ \epsilon_{x_M} [X_M] \} \cup \{ \mathbb{P} [X_v | X_{\text{Pa}(v)}] : e = (\{v\}, e^{\text{in}}), v \in \mathcal{U}, v \notin T \} \rangle_{[\emptyset]} \\ &\quad \cdot \langle \{ \epsilon_{x_T} [X_T], \epsilon_{x_M} [X_M] \} \cup \{ \mathbb{P} [X_v | X_{\text{Pa}(v)}] : e = (\{v\}, e^{\text{in}}), v \notin \mathcal{U}, v \notin T \} \rangle_{[X_O]} . \end{aligned}$$

We compare this contraction with the analogous decomposition

$$\begin{aligned} \mathbb{P}^c [X_O, D_T = x_T, X_M = x_M] &= \langle \{ \mathbb{P} [X_T = x_T | X_{\text{Pa}(T)}], \epsilon_{x_M} [X_M] \} \cup \{ \mathbb{P} [X_v | X_{\text{Pa}(v)}] : e = (\{v\}, e^{\text{in}}), v \in \mathcal{U}, v \notin T \} \rangle_{[\emptyset]} \\ &\quad \cdot \langle \{ \epsilon_{x_T} [X_T], \epsilon_{x_M} [X_M] \} \cup \{ \mathbb{P} [X_v | X_{\text{Pa}(v)}] : e = (\{v\}, e^{\text{in}}), v \notin \mathcal{U}, v \notin T \} \rangle_{[X_O]} . \end{aligned}$$

and notice that both differ only in a factor. Whenever $\mathbb{P} [X_T = x_T, X_M = x_M] \neq 0$ this factor does not vanish. After normalization with X_T and X_M incoming, the tensors are therefore equivalent. $\square$

Lemma 1.5 *When X_M does not contain a descendant of X_T then*

$$\mathbb{P}^c [X_M | D_T] = \mathbb{P} [X_M] \otimes \mathbb{I} [D_T] .$$

Proof. We causally augment the Bayesian network by the do variable D_T , and notice that also with respect to the causally augmented hypergraph, X_M does not contain a descendant of X_T nor D_T . We then investigate for arbitrary $d_T \in [m_T + 1]$ the contraction

$$\langle \{ \mathbb{P} [X_k | X_{\text{Pa}(k)}] : k \in [d], k \neq 0 \} \cup \{ \mathbb{P} [X_T | X_{\text{Pa}(T)}], \epsilon_{d_T} [D_T] \} \rangle_{[X_M]} .$$

We enumerate the nodes by a topological order, such that any node having X_M as a descendant is ordered before those nodes not having X_M as a descendant. For some $k \in [d]$ we then have that $X_{[k]}$ are the variables having X_M as a descendant and X_M itself. Lem. 1.1 implies for $\tilde{k} = 0$ and $\mathcal{U} = \emptyset$ that

$$\langle \{ \mathbb{P} [X_k | X_{\text{Pa}(k)}] : k \in [d], k \neq 0 \} \cup \{ \mathbb{P} [X_T | X_{\text{Pa}(T)}], \epsilon_{d_T} [D_T] \} \rangle_{[X_M]} = \langle \mathbb{P} [X_{[k]} | \emptyset] \rangle_{[X_M]} = \mathbb{P} [X_M] .$$

$\square$

Theorem 1.2 (Backdoor Criterion) *Let X_M satisfy the backdoor criterion with respect to X_T and X_O , no node in X_M be a descendant of X_T , and let $x_T \in [m_T]$ be such that for any $x_M \in [m_M]$ we have $\mathbb{P} [X_T = x_T, X_M = x_M] \neq 0$. Then we have*

$$\mathbb{P}^c [X_O | D_T = x_T] = \langle \mathbb{P} [X_O | X_T = x_T, X_M], \mathbb{P} [X_M] \rangle_{[X_O]} .$$

Proof. We have by the Bayes rule

$$\mathbb{P}^c [X_O | D_T = x_T] = \langle \mathbb{P}^c [X_O | X_M, D_T = x_T], \mathbb{P}^c [X_M | D_T = x_T] \rangle_{[X_O]} .$$

We apply Lem. 1.4 for arbitrary $x_M \in [m_M]$ and get

$$\mathbb{P}^c [X_O | X_M = x_M, D_T = x_T] = \mathbb{P} [X_O | X_M = x_M, D_T = d_T] .$$

By assumption of the backdoor criterion, X_M is not a descendant of X_T . Lem. 1.5 thus implies

$$\mathbb{P}^c [X_M | D_T = x_T] = \mathbb{P} [X_M] .$$

We conclude that

$$\mathbb{P}^c [X_O | D_T = x_T] = \langle \mathbb{P} [X_O | X_T = x_T, X_M], \mathbb{P} [X_M] \rangle_{[X_O]} .$$

$\square$

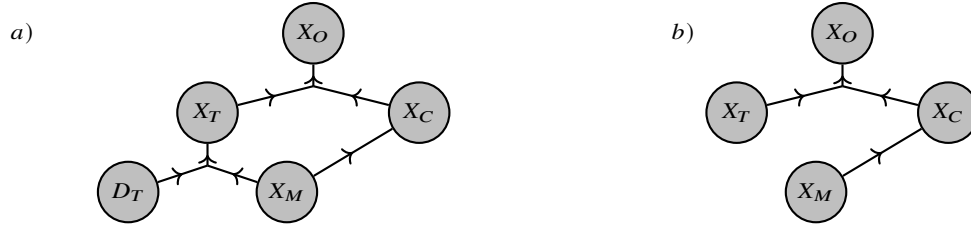


Fig. 1.6 Causal graph for the Backdoor Criterion (see Def. 1.3). a) The mediator variable X_M blocks any backdoor path from X_T to X_O , potentially through a confounding variable X_C . b) The effect of the intervention on X_T reduces the Bayesian network effectively to the shown subgraph.

1.2.3.2 Frontdoor Criterion

We now state the frontdoor criterion, see [?].

Definition 1.4 A frontdoor path from X_T to X_O is any directed path from X_T to X_O . A set X_M of variables satisfies the frontdoor criterion with respect to X_T and X_O , if the following hold:

- Any frontdoor path from X_T to X_O contains a node in X_M .
- There is no backdoor path from X_T to X_M .
- All backdoor paths from X_M to X_O are blocked by X_T .

Lemma 1.6 (Frontdoor Criterion) When X_M satisfies the frontdoor criterion with respect to X_T and X_O , then

$$\mathbb{P}^c[X_O|D_T = x_T] = \left\langle \mathbb{P}[X_M|X_T = x_T], \left\langle \mathbb{P}[X_O|X_M, X_T], \mathbb{P}[X_T] \right\rangle_{[X_O, X_M]} \right\rangle_{[X_O]}.$$

Note that the inner contraction has X_T as a closed variable.

Proof. We expand

$$\mathbb{P}^c[X_O|D_T = x_T] = \left\langle \mathbb{P}^c[X_M|D_T = x_T], \mathbb{P}^c[X_O|X_M, D_T = x_T] \right\rangle_{[X_O]}$$

Since there are no backdoor trails from X_T to X_M , we apply the backdoor simplification by Thm. ?? to get

$$\mathbb{P}^c[X_M|D_T = x_T] = \mathbb{P}[X_M|X_T = x_T].$$

Since X_M blocks all directed paths from X_T to X_O we have for any $x_M \in [m_M]$

$$\mathbb{P}^c[X_O|X_M = x_M, D_T = x_T] = \mathbb{P}^c[X_O|D_M = x_M].$$

By assumption, X_T further satisfies the backdoor criterion with respect to X_M and X_O . We can therefore apply the backdoor simplification by Thm. ?? again and get

$$\mathbb{P}^c[X_O|D_M = x_M] = \left\langle \mathbb{P}[X_O|X_M, X_T], \mathbb{P}[X_T] \right\rangle_{[X_O]}.$$

Combining these contraction identities, we arrive at the claim. $\square$

1.3 Counterfactuals

Following the formalism of Chapter 21 in [?], counterfactual queries are modeled as intervention queries on counterfactual twin networks. Counterfactual twin networks are two copies of the original causal model, one representing the factual world and one representing the counterfactual world. They further share response variables, which capture

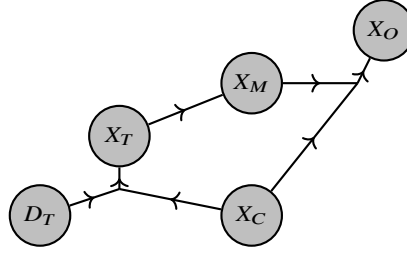


Fig. 1.7 Typical hypergraph, where X_M satisfies the frontdoor criterion with respect to X_T and X_O .

the functional relationships between variables in the causal model. These response variables are modeled as selection variables in the treason formalism.

Example 1.5 (Patient counterfactual query) Let us assume, that we know whether a patient did not get assigned to the treatment group, and we know whether he did improve. We want to ask, whether he would have improved, if he had been assigned to the treatment group. This is not an intervention query, but a counterfactual query, since we reason about a different outcome, although we already know that the patient has not been treated.

The causally augmented Bayesian Network is not rich enough to answer such queries. Let us consider a situation where X_T is chosen uniformly from $[2]$. There might be situations, where the probabilistic (and intervention) queries are the same, but the counterfactual queries differ:

- (i) The outcome is independent of the treatment.
- (ii) With probability 0.5, the treatment always leads to improvement and missing the treatment not. And also with probability 0.5, the treatment never leads to improvement, but missing the treatment does.

Intuitively, in the situation (i), the counterfactual query should return probability 1/2 for improvement under treatment. In the situation (ii), the counterfactual query returns the same value with probability 1 as observed in the real world. Both cases are however indistinguishable by causal models itself, since in both cases the causally augmented conditional probabilities are

$$\mathbb{P}^c[X_T|D_T] = \frac{1}{2}\mathbb{I}[X_T] \otimes \epsilon_2[D_T] + \left(\sum_{d_T \in [2]} \epsilon_{d_T}[D_T] \otimes \epsilon_{d_T}[X_T] \right)$$

and

$$\mathbb{P}^c[X_O|D_O, X_T] = \frac{1}{2}\mathbb{I}[X_O, X_T] \otimes \epsilon_2[D_O] + \left(\sum_{d_O \in [2]} \epsilon_{d_O}[D_O] \otimes \epsilon_{d_O}[X_O] \otimes \mathbb{I}[X_T] \right).$$

To answer counterfactual queries, we therefore have to also reason about the mechanisms behind the causal relationships, which we do by introducing response variables in the following.

1.3.1 Response Variables

We introduce response variables L_k , such that X_k has a deterministic dependence on L_k and the parents of a variable to the variable itself. The dependence is captured by a function

$$q_k : [p_k] \times \left(\bigotimes_{\tilde{k} \in \text{Pa}(k)} [m_{\tilde{k}}] \right) \rightarrow [m_k].$$

The response augmented conditional probability core is defined as (see Figure 1.8):



Fig. 1.8 Introduction of response variables: a) as a Bayesian network, b) decoration by the basis encoding of the corresponding function.

$$\mathbb{P}^f [X_k | L_k, X_{Pa(k)}] = \beta^{q_k} [X_k, L_k, X_{Pa(k)}]$$

Example 1.6 (Response variables for a random walk)

Let us investigate a specific example of a Markov chain, namely a random walk on the set $[m]$. We assume that with a probability $p \in [0, 0.5]$ we move to the right neighbor, also with p to the left neighbor and with $(1 - 2p)$ stay at the current state.

$$\begin{aligned} \mathbb{P}[X_k | X_{k-1}] &= p \cdot \sum_{x \in [m]} \epsilon_{x-1} [X_k] \otimes \epsilon_x [X_{k-1}] \\ &\quad + (1 - 2p) \cdot \sum_{x \in [m]} \epsilon_x [X_k] \otimes \epsilon_x [X_{k-1}] \\ &\quad + p \cdot \sum_{x \in [m]} \epsilon_{x+1} [X_k] \otimes \epsilon_x [X_{k-1}] \end{aligned}$$

Here we use the summation modulo m , that is $0 - 1 = m - 1$ and $m - 1 + 1 = 0$.

We introduce to each $k \in [d]$ a response variable L_k with dimension 3

$$\mathbb{P}[L_k] = \begin{pmatrix} p \\ 1 - 2p \\ p \end{pmatrix} [L_k]$$

where the zeroth coordinate is interpreted as the move to the next lower state, the first as staying at the position and the second as moving to the next larger state.

This interpretation is encoded in the function

$$q_k(x_{k-1}, \ell_k) = \begin{cases} x_{k-1} - 1 & \text{if } \ell_k = 0 \\ x_{k-1} & \text{if } \ell_k = 1 \\ x_{k-1} + 1 & \text{if } \ell_k = 2 \end{cases}$$

which has the basis encoding

$$\begin{aligned} \beta^{q_k} [X_k | X_{k-1}, L_k] &= \left(\sum_{x \in [m]} \epsilon_{x-1} [X_k] \otimes \epsilon_x [X_{k-1}] \right) \otimes \epsilon_0 [L_k] \\ &\quad + \left(\sum_{x \in [m]} \epsilon_x [X_k] \otimes \epsilon_x [X_{k-1}] \right) \otimes \epsilon_1 [L_k] \\ &\quad + \left(\sum_{x \in [m]} \epsilon_{x+1} [X_k] \otimes \epsilon_x [X_{k-1}] \right) \otimes \epsilon_2 [L_k] . \end{aligned}$$

By construction we have the decomposition

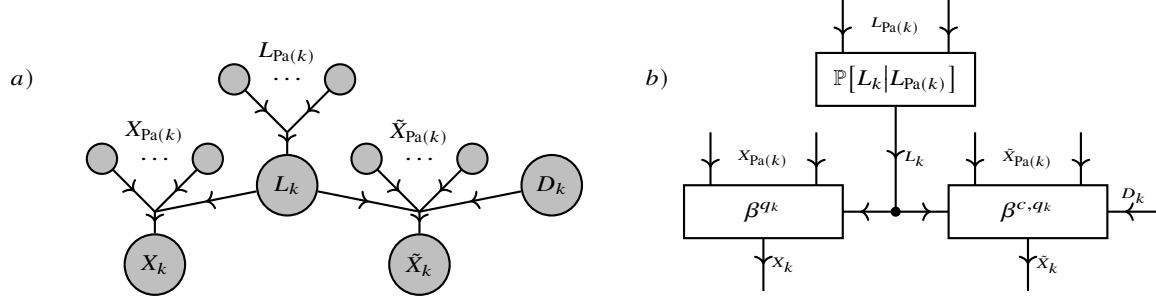


Fig. 1.9 Counterfactual twinned network: a) as a Bayesian network. b) Decorations by causal augmented basis encodings by a do-variable for X_k , a conditional probability distribution for L_k and a basis encoding for the counterfactual variable $\tilde{X}_k$.

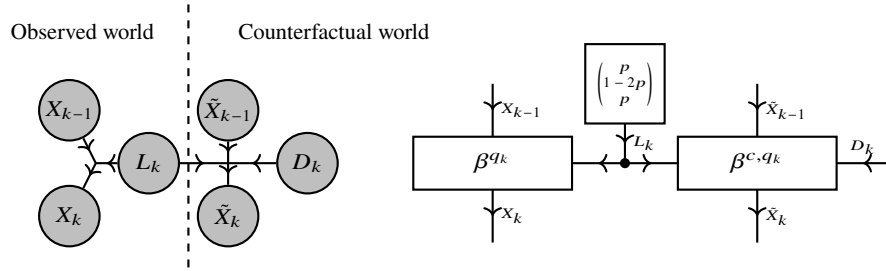


Fig. 1.10 Part of the counterfactual twinned network for the random walk (see Example 1.6 and Example 1.7).

$$\begin{array}{c} \uparrow X_k \\ \boxed{\mathbb{P}[X_k | X_{k-1}]} \\ \downarrow X_{k-1} \end{array} = \begin{array}{c} \uparrow X_k \\ \boxed{\beta^{q_k}} \xrightarrow{L_k} \boxed{\mathbb{P}[L_k]} \\ \downarrow X_{k-1} \end{array} .$$

1.3.2 Counterfactual Twinned Network

To answer counterfactual queries, we now construct the counterfactual twinned network (see Figure 1.9).

Definition 1.5 Let there be a response augmented Bayesian Network on a directed acyclic hypergraph $\mathcal{G} = ([d], \mathcal{E})$ with response variables $L_{[d]}$. Further another hypergraph $\tilde{\mathcal{G}}$ and a Bayesian Network on $\tilde{\mathcal{G}}$ modelling the joint distribution of the response variables $L_{[d]}$. Be build counterfactual copies $\tilde{X}_k$ of each variable X_k , which are controlled by the same response variables L_k as the factual variables X_k . The counterfactual twinned network is then the tensor network consists of the causal and response augmented conditional probability cores to $X_{[d]}$, the conditional probability cores of the response variables $L_{[d]}$ and the response augmented conditional probability cores to the counterfactual variables $\tilde{X}_{[d]}$.

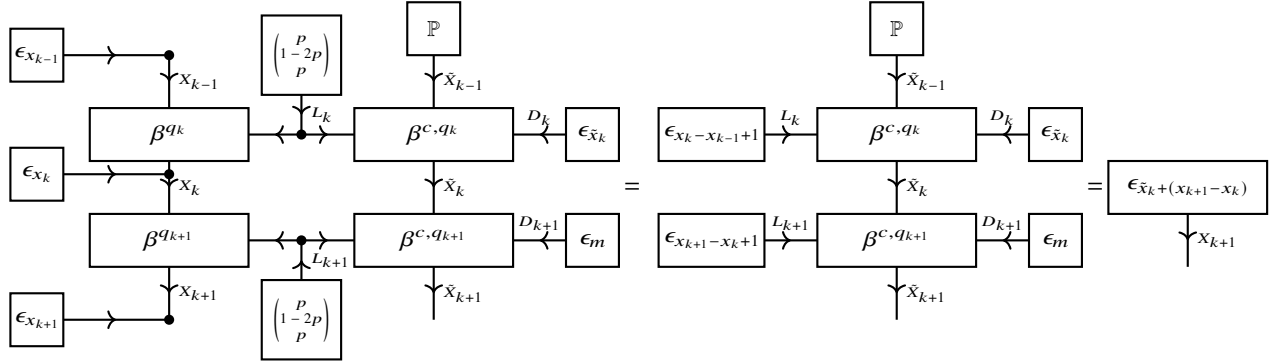
Example 1.7 (Counterfactual query for a random walk)

We build a counterfactual twinned network (see Figure 1.10), by adding counterfactual variables $\tilde{X}_{[d]}$ linked to the same response variables. Further, we include do variables to the counterfactual world, to answer counterfactual queries.

Setup:

- The variables X_{k-1}, X_k, X_{k+1} have been observed.
- The counterfactual variable $\tilde{X}_k$ is intervened on.

This counterfactual query is answered by the contraction



Let us consider an arbitrary set $\mathcal{U} \subset [d]$ of variables intervened on. For each $\tilde{k}$ we denote k as the previous variable which has been intervened on.

The counterfactual world is by the above contraction only supported at the indices

$$\tilde{x}_{\tilde{k}} = \tilde{x}_k + (x_{\tilde{k}} - x_k) .$$

If no intervention has been done before $\tilde{k}$, we define $\tilde{x}_k = x_k$ and have $\tilde{x}_{\tilde{k}} = x_{\tilde{k}}$.

Thus, the same steps (left, right neighbor or keeping positions) are done in the counterfactual world. Given the observed world, the distribution of the counterfactual world variables is therefore deterministic.

1.4 Learning

Given a parametrization of hypothesis distribution by the states of selection variables, we can include the intervention variables. The cores to be trivialized by intervention can depend on the state of the selection variables. For example, a selection variable might select whether the cores direction is from X_0 to X_1 or in the other direction. In both cases only one of the to intervention variables D_0 and D_1 influences the core.